Anurag Chattopadhyay
IT 4th Year
Roll: 9 (12200222024)

Speech & Natural Language Processing
PEC-IT 801C

Assignment - 1

Date - 28/01/2026

1) * Splits the raw text into smaller units called tokens.
* Fundamental initial step in text preprocessing ~~pipl~~ pipelines.
* Converts unstructured strings into structured data streams.
* Handles whitespace, punctuation, and special character separation.
* Enables vocabulary creation for machine learning models.
* Facilitates features extraction methods like Bag-of-Words.
* Helps identify sentences boundaries within large corpora.
* Supports downstream tasks like parsing and tagging.
* Can operate at word, character, or subword levels.
* Removes noise to standardize input for analysis.

2) * Syntax governs grammatical structure and word arrangement.
* Semantics focusses on the literal meaning of words.
* Pragmatics deals with context-dependent ~~mean~~ meaning and intent.
* Syntax ensures sentences follow structural language rules.
* Semantics interpret meaning independant of the situation.
* Pragmatics considers the speaker's purpose and audience.
* Syntax handles rules like subject-verb agreement.
* Semantics ~~ref~~ resolves ambiguity in specific word definitions.
* Pragmatics helps interpret sarcasm, irony, and metaphors.

3) * Instance-based learning algorithm used for text categorization.
* Represents documents as vectors in a feature space.
* Calculates similarity between test and training documents.
* Uses metrics like Cosine similarity or Euclidean distance.
* Identifies k-closest neighbours to the input.
* Assigns class based on majority vote of neighbours.
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